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Original article

Aggressive behavior: A language to be understood

Les comportements agressifs : un langage à comprendre

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ABSTRACT

Aggressive behaviors could be considered as a dynamic of communication, in which aggression is a language to be understood, to be deciphered by two protagonists : the aggressor coping with a stressful and threatening situation and the aggressed individual coping with an aggressive acting out. The following questions are addressed: (a) What does aggression mean to the aggressor, what does it mean to the aggressed individual? (b) What does the aggressor want or try to express, and why does he or she use this mode of expression and action over another? (c) How does the aggressed individual react, and what is the impact of his or her response on the aggressor? This article reviews studies on the definition of aggression, its measurement, its developmental role and its associated risk factors in children and adolescents. First, aggression in children and adolescents with typical and atypical development is examined based on a developmental approach, clinical case studies in child and adolescent psychiatry, and an empirical study on aggression in autism. Then, in light of these studies, the problem of violence among younger and younger children and adolescents, is discussed. Finally, taking together these studies and discussion, a model is proposed that accounts for factors involved in aggressive behaviors and for different possible levels of intervention. Physical and/or psychic threats generate stress and may therefore lead to offensive or defensive aggression. Stress has to be considered in its physiological dimension (biological stress responses) and psychological dimension (perceived stress). This model reveals a vicious circle: when the main response to aggression is repression, it may reinforce physical/psychic threats and stress perceived by the aggressor, and in turn aggressive behaviors. Intervention can occur at three levels: the stressful situation, the aggressor (perception of the stressful situation and his or her response), and the aggressed individual (perception of the aggression and his or her response). This model, like all models, is limited but it offers a discussion and perspectives to understand the different links that form the “chain of aggressive behavior” and the relationships between exposure to violence and expression of violence. The article concludes on the interest of a multidisciplinary approach to aggression integrating physiological, psychological and sociological dimensions.

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R É S U M É

Les comportements agressifs peuvent être considérés comme faisant partie intégrante d'une dynamique de communication, dans laquelle l'agression est un langage à comprendre, à décrypter entre deux protagonistes : la personne qui agresse face à une situation stressante et menaçante (menace physique et/ou psychique), et la personne agressée qui fait face à un passage à l'acte agressif. Les questions suivantes sont ici soulevées : (a) Que signifie l'agression pour la personne qui agresse ? (b) Que veut-elle ou essaie-t-elle d'exprimer au travers de la conduite d'agression, et pourquoi utilise-t-elle ce mode d'expression

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et d'action plutôt qu'un autre ? (c) Comment la personne agressée réagit-elle, et quel est l'impact de sa réponse sur la personne qui agresse ? Cet article est une revue de plusieurs études portant sur la définition de l'agression, son évaluation, son rôle développemental, et les facteurs de risque ou circonstances susceptibles de contribuer à la survenue de conduites d'agression chez les enfants et adolescents. Tout d'abord, les comportements agressifs chez les enfants et adolescents au développement typique et atypique sont examinés au travers d'une approche en psychologie du développement, mais aussi de cas cliniques en psychiatrie de l'enfant et de l'adolescent, et d'une recherche empirique sur les comportements hétéroagressifs dans le Trouble du Spectre de l'Autisme. À la lumière de ces études, le problème de la violence observée chez des enfants et adolescents de plus en plus jeunes, est discuté. Enfin, à partir de ces études et de cette discussion, un modèle est proposé permettant de rendre compte de facteurs impliqués dans les comportements agressifs et de différents niveaux possibles d'intervention. Les menaces physiques et/ou psychiques sont génératrices de stress et peuvent par conséquent déclencher une conduite d'agression offensive ou défensive. Le stress provoqué est à appréhender tant dans sa dimension physiologique (réponses biologiques au stress) que dans sa dimension psychologique (stress perçu par l'individu). Ce modèle révèle un cercle vicieux : quand la principale réponse à l'agression est la répression, cela risque de venir renforcer les menaces physiques/psychiques et le stress perçus par l'auteur de l'agression, et par conséquent ses comportements agressifs. Les interventions peuvent être réalisées à trois niveaux différents : le niveau de la situation stressante, le niveau de la personne qui agresse (perception de la situation stressante et sa réponse), et le niveau de la personne agressée (perception de l'agression et sa réponse). Ce modèle, comme tous les modèles, est réducteur mais il offre une discussion et des perspectives de compréhension des différentes associations qui se succèdent en formant la « chaîne du comportement agressif », compréhension notamment des relations entre l'exposition à la violence et l'expression d'une violence. Cet article conclut sur l'intérêt d'une approche multidisciplinaire des comportements hétéroagressifs intégrant les dimensions physiologique, psychologique et sociologique.

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1. Introduction

We are all confronted with aggressive behavior and the problem of violence, be it in the field of psychiatry or in our everyday lives. Few behaviors have interested as varied a group of specialists (psychiatrists, psychologists, sociologists, neurobiologists, geneticists, ethologists).

The first problem is to define aggression. Aggression has been defined as a behavior deliberately aimed at inflicting physical damage to persons or property [1]. Assessment tools used currently in psychiatry to measure aggression are based on diverse definitions of aggression. For some, it is defined by uncontrollable anger. For others, objects have to be destroyed. Still others require physical aggression toward people. The term aggression does not even appear in the glossary of the DSM-5 [2]. Buss [3] defined aggression as a response that delivers noxious stimuli to another organism. The noxious stimuli may be physical (e.g., hitting, punching, stabbing or shooting) or verbal (e.g., cursing or threatening). But already, there is a problem of interpretation and communication: what is noxious for some people is not perceived and understood as noxious by other people. Other authors consider that anger is one of the main factors of aggression and therefore use anger scales or angry feeling items in order to measure aggression [4–6]. Thus, it seems that the measurement of aggression is greatly influenced by the specific definitions and connotations ascribed to it, such that different instruments may measure different factors of aggression [7]. Concerning the nonstructured assessments of aggression, many of them are based on criminal records [8–10]. But equating aggression with criminality is problematic because crime implies a legal definition and what is a crime in a certain society or country is not recognized as a crime in another one. For example, in the French judicial system, crime is legally defined in the following terms: “all offenses that laws punish with by afflictive or infamous penalty (as opposed to minor infractions or misdemeanors)” [11]. In the United States, crime includes burglary, car theft, and all other sorts of crimes against property. The legal definition of crime thus depends on the social and historical context in which

laws are made. In the same way, the very notion of aggression differs according to the individual aggressed and his/her sociocultural surroundings. And even when an aggressive behavior is recognized as such, its connotations vary depending on the sociocultural environment. For example, aggressiveness has positive connotations among American businessmen, as in the image of a fighter. In China, however, women in particular are duty bound not to externalize their aggressiveness. What is the difference, the limit, between a normal aggressive behavior and a pathological one? Certain societies or individuals consider aggression a normal survival behavior, whereas others consider it a dangerous, maladjusted behavior, which takes aim at inappropriate targets. What is “maladjusted” in one society can be “adjusted” in another. This introduction allows us to appreciate the complexity of aggressive behaviors and the necessity of taking into account the sociocultural context in which these behaviors occur.

We thus find ourselves in a dynamic of communication, in which aggression is a language to be understood, to be deciphered by two protagonists: the aggressor and the aggressed individual.

The objective of this article is not to present a synthesis of the entire study of aggression but rather, to focus on the following questions: What does aggression mean to the aggressor? What does it mean to the aggressed individual?

What does the aggressor want to express, and why does he/she choose this mode of expression and action over another? How does the aggressed individual react, and what is the impact of his/her response on the aggressor?

First, aggression in children and adolescents with typical and atypical development will be examined based on a developmental approach, clinical case studies in child and adolescent psychiatry, and an empirical study on aggression in autism spectrum disorder (ASD). Then, in light of these studies, both the problem of violence among younger and younger children and adolescents and the response of the justice system, will be discussed. Finally, taking together these studies and discussion, a model will be proposed that accounts for factors involved in aggressive behaviors as well as for different levels of intervention.

2. Aggression in children and adolescents with typical and atypical development

2.1. Developmental approach

The field of developmental psychology sheds some light onto the evolution of aggression in the first years of life and thus its possible functions. Mayes and Cohen [12] highlight one specific function of aggression in early development – the role of aggression in the child's moves toward separation and individuation described previously by Winnicott [13] or Mahler, Pine, and Bergman [14]. Therefore, it could be important not to try to suppress a behavior, which is involved in the process of separation-individuation. Aggression makes it possible for the child to be a separate individual who is then capable of the identifications, introjections and partial merging so characteristic of mature object relations and love relationships [15,16]. Aggressive behaviors, and more precisely physical aggression, reach a peak of frequency when typically developing children are between 18 and 24 months of age, and start then to decrease from the age of 2 years old concurrently with the development of verbal language [17–21]. In agreement with this observation of relationships between verbal language and aggressive behaviors, the Dominick study did not find any significant differences in frequency of aggressive behaviors between children with ASD involving verbal communication impairments and children with history of language impairment but without ASD [22]. In the same line, severe social communication impairments observed in children and adolescents with autism and/or intellectual disability, including an absence or lack of verbal language as well as a deficit of empathy and theory of mind (i.e., the difficulty to represent self's or other's thoughts and emotions), may explain the occurrence and persistence of aggressive behaviors due to problems for these individuals to express their emotions (anger, fear, etc.) or frustration, and release their stress (physiological and psychological stress) [23–26]. Also, it is noteworthy that verbal aggression increases between age 2 and 4 in typically developing children [27]. All types of aggressive behaviors decrease markedly as the child approaches age 5 [28,29]. Hartup [30] found that 4- to 6-year-old children were more aggressive than 6- to 7-year-old children, primarily due to the younger children's tendency to use aggression to acquire or reclaim desired objects.

By school age, aggressive behaviors are reduced, mainly due to inhibition of aggression resulting from social learning, and are more often a response to an insult or other threat to self-esteem. Persistent aggression would result more from a lack of “learning not to aggress” rather than from an excess of “learning to aggress” [31–33]. More precisely, inhibition of aggressive behavior results from learning to control or regulate emotions (such as anger), impulsivity and acting out through education in family, school, social and societal environments. This requires the existence, communication and application of educative rules (parental, school, social and societal rules) in the group, and their integration and respect in the individual. The persistence of aggressive behaviors in adolescence and adulthood questions a lack of limits set usually by a structured environmental framework as well as a failure of inhibitory processes. Developmental inhibitory processes of aggressive behavior exist both in humans and other social animals, allowing the survival of the group that could be a real issue if the individuals of this group were spending their time to aggress each other. When a young Rhesus monkey displays recurrent aggressive behavior, this youth is excluded temporarily from the group and stays alone until the monkey is able to calm down [34]. The need, for humans and other social animals, to be part of the group and to keep their place within the group, is essential (including for the individual's development and survival). This fundamental need is related to the very early development of social self

(around 5 months of age) which appears before the recognition in the mirror of others (occurring around 8 months of age) or one-self (occurring around 2 years of age), and may correspond to an archaic process necessary to the survival of the individual [35]. The risk to be excluded from the group or the experience of a transitory exclusion leads the individual to control his or her episodes of anger and aggressive behavior as well as find compromises and new strategies for conflict resolution. However, when exclusion from the group is prolonged with long-term isolation, it can be perceived by the individual as a very violent and stressful situation, and therefore increase in response his or her aggressive behavior in a real vicious circle with a risk for desocialisation. Many studies in animal models have reported since several decades aggression triggered by stressful situations/stimuli and also increased aggression resulting from long-term social isolation-induced stress [36–40]. Prolonged exclusion with social isolation can even lead in Rhesus monkeys excluded from the group for aggressive behavior to their death given that they cannot survive a longtime out of the group [34].

2.2. Clinical case studies in child and adolescent psychiatry

Aggressive behaviors were found to be two to three times more prevalent among children with developmental disabilities compared to typically developing children [41]. Furthermore, McClintock [42] reported that autism was a “risk marker” for aggression and disruptions to the environment when compared to non-autistic individuals with developmental disabilities. Aggressive behaviors in children and adolescents with autism limit their integration in community and educational activities, and can impact their prognosis when they are recurrent, persistent and/or uncontrolled [43–47].

Today is the case study of John, a seven-year-old child with autism. His caregivers explain the problems that they have had with him over the past few weeks. In short, John is becoming more and more aggressive. He bites and scratches them, explodes into fits of aggression, and has sudden violent reactions to the point where they are starting to be afraid to get near him. The child psychiatrist then takes his pen, and prescribes a sedative antipsychotic. It's an all too familiar story. In some child psychiatry in-patient departments, the chemical treatment of aggressive behaviors in autism spectrum disorder has a large following.

It seems, however, that the appearance of aggressive behaviors in individuals with autism spectrum disorder could also represent a positive evolutionary phase. Several French clinical teams report that cases in which children with autism pass from non-aggressiveness (absence of externalized aggressive behavior associated with severe autistic withdrawal) to aggressive behaviors directed against others (hitting, biting and scratching adults or other children) are actually cases of positive evolution [48].

What is the significance of aggressive behavior in a child with ASD who has no verbal language? Indeed, the appearance of aggressive behaviors directed against others bears witness to the child's growing interest in the outside world, to his or her ability to emerge from autistic withdrawal, be connected with others, and enter into relationships (however aggressive) with others. Aggression can be understood as of a way for the individual to express him or herself and therefore as part of social communication.

Aggressive behaviors in children with ASD are most often triggered by an anxiety-producing stimulus. Stress appears to be a fundamental factor in the onset of these behaviors (self-injurious behavior as well as aggressive behavior directed against others) [49–51]. Aggression can occur at the mere perception of such a stimulus as, for example, the popping of a balloon. The trigger can also be frustration or, moreover, fear inspired by the child's unfamiliar exterior surroundings.

Obviously, we have to protect ourselves from aggression. But there is a difference between, one hand, containing violence to protect ourselves (by physical or chemical means) at the moment at which the violence occurs, while maintaining a dialogue, and, on the other hand, chronically repressing it using an anonymous intermediary (medication or isolation prescribed over several months). In the first case, we sustain the dynamic of communication between the aggressor and the aggressed individual. In the second case, communication is interrupted by the long-term chemical or physical isolation of the aggressive individual who is not anymore able to express him or herself. The absence of communication and the chronicity of repression are what make this case negative. At the moment of aggression, a physical restraint, an injection of sedatives or a short-term isolation, can actually reassure the patient. It offers the patient a soothing period by holding him or her physically (from tactile holding with body contact/envelop, such as embracing tightly the individual like a mother does with her upset child, to short-term physical restraint), chemically (sedative medication) and/or in a calm space (transitory isolation). Also, it sets limits and shows the patient that we are able to protect ourselves and that he or she cannot destroy us. (“destruction” can in fact be very distressing for the aggressor). Finally, it enables us both to calm the patient, to divert him or her from the acting out, and then a verbal dialogue more constructive can ensue later.

One of the most telling examples of psychiatric repression is the case of a young man for whom I was called one night when I was on duty. I was asked to come to the psychiatric ward to confirm the accidental death of a 24-year-old psychotic patient. I found this man lying dead on the ground, near his mattress, entirely naked, in an isolation room, which he had lived in for more than three years. Seeing my surprise, the nurse explained to me that he had been isolated after he tried to strangle the director of the psychiatric department. No clothing, furniture, or utensils were allowed in this room due to self-injurious behaviors of the patient. The only contact he had with other human beings were the times he was brought meals on a tray left on the floor of his room, which he ate with his hand. Little by little, this man had totally regressed. He had lost all language, had become more and more aggressive, and lived like an animal, to the point where no one even tried to approach him. As indicated previously in this article, many studies have described the reinforcement of aggression in male mice and rats that follows long-term social isolation-induced stress [36–39,52,53]. This man, over a three-year period of isolation, was reduced to living—and dying—like an animal. This story did not take place in some faraway country or even at some faraway time. This happened in France, in the 1990's.

The role of the psychiatrist should be, as it is for all physicians, to care for the patient. In reality, the psychiatrist often thinks only of limiting aggression by eliminating potentially dangerous behavior or even, as in the case above, the source of the danger itself. Most of the time we do not try to understand what the aggressive behavior means, or what the patient tries or wishes to express through this behavior. It is, of course, easier to tackle symptoms, to make a troublesome behavior or person “disappear” for reasons of protection and security, than to make ourselves try to understand.

We protect ourselves from aggression, which is a “statement”, with chemical intervention or physical isolation, which are equally “statements”. However, it is possible to converse, to verbalize at a certain distance from aggression. Psychiatric nurses and educators are sensitive to this, perhaps because they spend a lot of time with patients.

Laurent is a 17-year-old adolescent with autism who has become brutally aggressive. He bites, scratches, and hits other children and staff for no reason. This behavior is all the more disturbing because of Laurent's considerable height and strength. His caseworkers thus tried to understand the meaning of his

aggression. They made a connection between the appearance of Laurent's aggressive behavior and the recent pregnancy of his mother. They tried to work through Laurent's aggression by talking with him about this connection. They even created a game in which Laurent displaced his aggression on a “pregnant” doll, which was to represent the real conflict. Laurent's aggressive behavior has decreased and totally disappeared several weeks later.

It is interesting to observe that tolerance of aggression within a clinical team grows when there is a better understanding of aggressive behavior. A case similar to that described above occurred at Benhaven, a center in Connecticut for individuals with autism. An adolescent with ASD became suddenly extremely violent. The nurses could no longer control him and started to be afraid of him. In order to try to understand the situation, one nurse traced and studied the patient's curve of daily aggression over several weeks, noting the number and type of aggressions. She thus made the connection between the outbreaks of aggressive behavior and problems that this patient was encountering in his work activities (adolescents with autism can achieve a certain level of social participation by working on small projects). The curve of aggression had two purposes: first, to give the team the feedback that it was possible to control better an apparent uncontrollable behavior, and thus to be less afraid of it. Second, to allow a better understanding of the aggressive behavior and therefore to intervene more effectively. The team indeed talked to the patient about what they thought was the source of his anxiety and was able to intervene directly on it. The teams at Benhaven get together regularly to discuss the problems they encounter with individuals with autism in order to understand better “difficult” behaviors like aggression.

Dealing with aggression is not a matter of taking thoughtless risks. When I was on call for the first time in Psychiatric Emergency, a man was brought in by the police strapped to a stretcher. My initial reaction was to tell the nurses to untie him. As soon as he was untied, he jumped up and punched me. I helped immediately the nurses get him back under control, put him back on the stretcher, and immobilize him. The police and the nurses were laughing. . . These same nurses taught me how to act in an isolation room with an aggressive or agitated patient. Lower your voice, which obliges the patient to be less agitated if he or she wants to hear you (the more you raise your voice, the more you risk exacerbating aggressive behaviors). Speak in a calm, reassuring, composed voice while not revealing any anxiety. Hunch your shoulders up with your back against the wall (this way the patient sees you as less dangerous and becomes less aggressive). It is necessary both to show that you are not afraid and not to look dangerous to the patient. You must neither intimidate nor be intimidated. Aggressive behavior can be a response to survival instincts and be in fact a defensive behavior (*defensive aggression*): “Attack before being attacked”.

It appears based on these clinical cases that aggression and anxiety/stress in the aggressor, and fear in the aggressed individual, are closely connected. However, these observations are derived from anecdotal reports and their conclusions need to be clarified and confirmed by empirical research studies, especially with regard the relationships between aggression and anxiety/stress in the aggressor and fear with different tolerance thresholds in the aggressed individual.

2.3. Research study on aggression in children and adolescents with autism spectrum disorder

2.3.1. Background

There are numerous anecdotal reports [54–56] of violence and aggression in children and adolescents by parents, caregivers and mental health professionals, such as the clinical cases presented in the section above (Section 2.2). The present study was conducted to understand better and characterize the different types of

aggression occurring in different observational situations (parental, caregivers, blood drawing) in a large sample of low functioning children and adolescents with autism spectrum disorder, and to compare aggression across the autism and typically developing control groups during the blood drawing situation. This study was part of a larger project (INSERM CRE 931009) that had as its overall objective the examination of associations between behavioral profiles and biological variables in children with autism. The larger project required a blood drawing and included, in addition to the study of aggression (other-injurious behavior), an examination of the relationships between self-injurious behavior, plasma endorphin and pain reactivity [51,57,58].

2.3.2. Methods

The study was conducted on 74 children and adolescents with autism and 115 typically developing control individuals matched for sex, age and pubertal stage. Children and adolescents with autism were recruited from French day-care facilities, and included 49 males and 25 females (mean age = 11.6 years, SD = 4.5; 32 pre-pubertal, 16 pubertal, 26 post-pubertal). Typically developing control individuals were recruited over a three-month period from the Reims preventive medical center, and included 75 males and 40 females (mean age = 12.7 years, SD = 5.9; 45 pre-pubertal, 27 pubertal, 43 post-pubertal).

Cognitive functioning of individuals with autism was assessed by two psychologists using the age-appropriate Wechsler intelligence scales and the Kaufman K-ABC [59]. All individuals with autism were cognitively impaired (mean full scale IQ $\pm$ SD: 42.2 ± 3.2 , with range of 40–58).

Assessments of other-injurious behavior in individuals with autism were performed using the Other-Injurious Behavior Scale (OIBS) [60,61]. The OIBS was used to assess other injurious behavior (OIB) by providing quantitative ratings (scored from 1 to 7) based on frequency, severity and duration for 15 types of OIB. This scale assesses the current and lifetime OIB. It provides also a qualitative context subscale indicating the circumstances in which the OIB occurs, the triggered factors, and the concurrent behaviors occurring just before the OIB in order to understand better the appearance of OIB. The OIBS has been previously found to have good discriminative capacity and to be reliable (inter-rater reliability) and valid (internal and external validity) for assessment of other-injurious behaviors in autism [61]. Aggression was assessed in three observational situations (parents at home, two caregivers at day-care, a nurse and a child psychiatrist during blood drawing) using validated scales.

2.3.3. Results

A high percentage of individuals with autism displayed OIB [caregiver evaluation: 43/74 (58%); parental evaluation: 25/74 (34%)], especially slapping, pinching-holding others tight and explosion-scattering of objects, but the parents observed significantly less OIB at home in their children (for any types of OIB) than the caregivers at day-care center ($P < 0.01$). According to the parental and caregiver observations, the most frequent emotional circumstances in which OIB occurred were frustration, anger and opposition. In addition, the most frequent concurrent behaviors reported to occur clearly just before the appearance of OIB in individuals with autism, were anxiety-related behaviors and excitation (including joyful state) according to the parental or caregiver observation. Furthermore, the frequency of OIB was significantly higher in individuals with autism compared to typically developing control individuals during the blood drawing (23% vs. 0%, $P < 0.01$).

2.3.4. Discussion

The different results between the parental and caregiver observational situations underline the key role of the environment and

relational context in assessing OIB and developing therapeutic perspectives. This relational context manifests itself (a) in the patient who modulates his/her behaviors as a function of the environment, i.e. the people who are around and the situation including the physical environment (location, furniture, etc.), and (b) in the observer who can perceive, interpret, and thus score differently OIB. In addition, the results suggest that in a stressful situation, such as the blood drawing, individuals with autism release their stress through behaviors such as OIB, whereas typically developing individuals can regulate and express their stress using verbal and non-verbal communication, social interaction, and other cognitive skills such as mental coping strategies, symbolization skills with representation and anticipation of the stressful situation. This hypothesis is strengthened by the present findings showing that the most frequent emotional circumstances in which OIB occur in autism are frustration, anger and opposition, and that the more frequent concurrent behaviors occurring just before OIB are anxiety-related behaviors and excitation. These results are in line with the aggression-frustration theory [62] stating that aggression increases when the individual is frustrated due to being blocked in reaching his or her goal, which creates anxiety and stress. Furthermore, it appears important to understand OIB in autism situationally, in the context of the relational dynamics arising between an individual expressing him or herself through a particular behavior, here the aggression, related to a situation, and a social environment that perceives this particular behavior and responds to it with different tolerance thresholds according to the aggressed individual.

3. Aggression in younger and younger children and adolescents

Violence is involving younger and younger children and adolescents. The Home Office research paper on youth crime in England and Wales in 2009/2010 shows that a quarter of all crimes are committed by offenders (mostly males) under the age of 18 and represent more than a million crimes in just one year [63]. Government statistics indicate that nearly one in five of all youth offenders first arrested went on to commit more than ten offences over the next nine years, with the youngest most likely to reoffend (youngsters are also likely to reoffend after completing custodial sentences); about three quarters return to crime within a year [63]. In France, minors represent 9.5% (234,000 children and adolescents) of all individuals involved in criminal cases (2.47 millions of individuals). In 2013, of the 234,000 minors involved in criminal cases, more than half (53%) were under 16-years-old (40% were between 13- and 15-years-old, and 13% were under 13-years-old) and were mainly boys (83%) [64]. Furthermore, the nature of the cases in which minors were involved differed from that of adults. Property offenses dominate with about half (49%) of minors involved. Offenses against persons (27%) constitute the second type of offenses committed by minors, including: 13% for minor violence, 6% for violence against minors, and 4% for more serious violence, such as sexual assault and rape (1%). Offenses against the government authority are quite frequent (5%), among which are petty theft and offenses related to carrying a weapon, as well as traffic offenses (4%). Judiciary responses to juvenile delinquency have become more systematic: in 1994, the judiciary response rate in cases involving minors was 60%, for 94% in 2013 [65].

Homicide consistently ranks in the top three leading causes of death among persons aged 10–24 years in the United States, after motor vehicle-related and other types of accidents (Center for Disease Control, Federal data); more precisely, approximately 12 young people are victims of homicide each day and almost 1400 cases are treated in emergency departments for nonfatal

assault-related injuries [66]. Homicide is the leading cause of death for African-American youth, the second leading cause of death for Hispanic youth, the third leading cause of death for American Indian/Alaska Native youth, and the fourth leading cause of death among White and Asian/Pacific Islander youth [66]. According to the World Health Organization (WHO) in 2020, an estimated 200,000 homicides occur worldwide among youth 10–29 years of age each year, making it the fourth leading cause of death for people in this age group. These 200,000 homicides represent 42% of the total number of homicides each year [67].

This violence and aggression might be “survival behaviors” due to a climate of insecurity and fear in which these children and adolescents grow up and in which they can easily access weapons. In New York, half of the high schools have weapons detectors. Students explain that they need guns in class to protect themselves and about 25% of American high school adolescents in 2017 recognize carrying a firearm at least once a month [68,69].

Violence has become a survival technique for the poor [70]. An example of this are the Chinese gangs in New York. These teenagers have arms, lawyers, and regular incomes at their disposal and operate in Chinatown by extorting merchants in exchange for protection that these same merchants pay to sometimes three or four different gangs. They look for potential recruits among the young Chinese immigrants. The profile of a typical recruit: a recently immigrated Chinese family who is in debt because of their move to the United States. The two parents work non-stop as do their children after school. Mantin [71] reported that more than half of the Chinese students at Seward Park High School worked after classes nearly 24 hours a week, and the clandestine businesses where these children work were numbered at no less than 500. The parents pressure their children. They must succeed in school, acquiring a sufficient education and a good knowledge of English, otherwise their future will be limited to working in factories or restaurants. Stress is all the greater because these children must adapt to a new environment, a new language (when their parents themselves do not speak English and therefore cannot help them), with the knowledge that their father is in a “less prestigious” profession than he had in China. In this context, failure in school can lead to profound family conflict. The teenager starts to cut classes and hang around Chinatown, becoming easy prey for the Chinese gangs who are always on the lookout for young recruits and who do not hesitate to use force. The first time, they may invite these children or adolescents to lunch, but if they are slow to decide, they are not afraid to use extortion or physical intimidation. In exchange, the gang makes it possible for them to have money (Chinese gang members are actually paid like professionals), to have guns and lawyers, to satisfy a need for recognition. . .

In a more general way, the children who are at risk are those who grow up in hostile environments which border on poverty, homelessness, drugs, alcohol, violence, and crime; those who face physical and sexual abuse; those who are raised by immature and still adolescent parents. And they affirm themselves by surviving this hostile environment. At eight years old, they may be well on their way to becoming drug dealers or murderers because of the all-powerful incentive of money. Some of these children, eight or nine years old, already sell drugs at school and can kill for a pair of tennis shoes or a jacket. . . An important aspect of the “drug culture” in poor communities (primarily black and Hispanic ghettos) is also a fundamental reason for its success: it is the only way for young people with no education to ever hope of having money, of achieving the “American dream”. Several articles have pointed out the resemblance between the enterprising spirit of the American capitalist and that of the drug dealer [72–74].

A study of 500 Afro-American preadolescents (drugs dealers and children who have committed assaults and other crimes) suggests that their main problem is a profound lack of self-worth [75]. These

children kill for things of virtually no value because they consider anything as being more valuable than themselves. If their own lives have no value, what can the lives of others mean to them? They have never experienced a warm environment or emotional support and therefore can only express their need to exist by antisocial behaviors. Children who deliberately engage in antisocial acts are often fearless and, consequently, less inhibited, less sensitive to punishment, and more sensitive to potential gains from their antisocial activities [76,77]. With psychologists or psychiatrists, they reveal no feeling: no fear, no guilt, no love. They are truly a “dehumanized generation” and a generation that is the result of a dehumanizing environment.

Most of the time, society responds to this violence with increased repression. It seems as if we need more and more prisons, police, and executions. And what about this law applied in certain states such as Connecticut: the age of majority is 18, but if a child of 14 commits a murder he or she is then considered an adult, is no longer subject to juvenile detention, and can be imprisoned or, like an adult, condemned to death? In fact, the very notion of adulthood in the United States is elusive. At 14-years-old, you are considered responsible for a crime if the crime is particularly serious, at 16 you can drive a car, at 18 you become a voting citizen, and it is only when you are 21 that you have the right to buy alcoholic beverages. These laws can also vary from one state to another. It would seem that the responsibility is based on the severity of the crime committed instead of on the very responsibility of the individual who commits it. So children or mentally ill committing crimes could be subject to the death penalty. Some people explain that these measures are preventive to protect society; they take youths who are “too dangerous” out of society (by means of prison or capital punishment). Furthermore, legally, crime could be considered as an act that allows a child or an adolescent to achieve adulthood; it is a “rite of passage” that enables this child/adolescent to become an adult. Such a system can only increase criminality among teenagers committing crimes in order to be considered gang leaders, in order to be “respected”, to gain access to adulthood, to have an identity (even criminal), just to be “someone”. These adolescents want to be recognized, valued, and heard, but the response they get to their acts are other acts: the legal gavel falling like a guillotine and incarceration. This may reinforce their need to be recognized, especially when there is no place to be heard, when people do not try to understand, but just to defend themselves. And what lies in store for these fourteen year olds who suddenly find themselves in adult prisons? In prison, they will meet with violence, real gang leaders, and drugs. When they get out, they will be jobless and thus will have to face as much psychological as financial insecurity and stress. It is noteworthy that repression by incarceration might be a risk for reinforcement of aggressive behaviors among these adolescents.

4. General discussion: a model of understanding aggressive behaviors

Taking together the studies developed in Section 2 and the discussion in Section 3 on aggression in younger and younger children and adolescents, a model is proposed featuring different levels of understanding and intervention for aggressive behaviors. This model, like all models, has limitations, but it allows us to understand certain factors and mechanisms involved in aggressive behavior (Fig. 1).

Physical or psychic threats produce anxiety (anticipatory) and stress (biological stress responses and perceived stress), and can engender offensive or defensive aggressions. Stress has indeed to be considered in its physiological dimension (biological stress responses) and psychological dimension (perceived stress by the individual) [78]. Concerning the stressful situation, it could be a

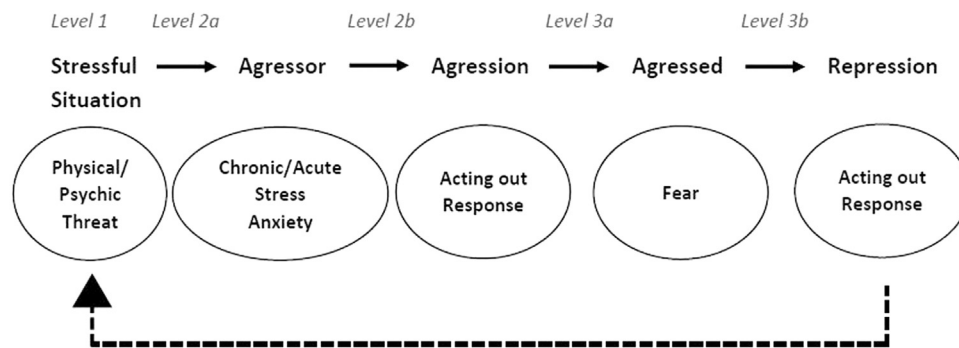


Fig. 1. Model of understanding aggressive behaviors. This model proposes different levels of understanding and intervention for aggressive behaviors. Physical and/or psychic threats generate stress and may therefore lead to offensive or defensive aggression. This model reveals a vicious circle: if the only or main response to aggression is repression, it may reinforce physical/psychic threats and stress perceived by the aggressor, and in turn aggressive behaviors. Interventions can occur at different levels: at the level of stressful situations (level 1), at the level of the aggressor (perception of the stressful situation: level 2a, and his or her response: level 2b), and at the level of the aggressed individual (perception of the aggression: level 3a, and his or her response: level 3b).

matter of a threat to territory or property. Territory and property are considered by the aggressor as an “extension” of himself or herself, so if they are encroached upon, then he or she can feel personally violated. Physical or psychic threats can also be stressful stimuli (like the popping of a balloon for a child with autism), frustration, a hostile and violent environment, a situation, which undermines one’s self-esteem or negates one’s existence or one’s needs. These threats generate acute or chronic stress, which is associated with physical or psychic pain. Psychic pain comes in various forms. It can be fear, anger, the loss of self-esteem, etc.

This model, as presented here, reveals a vicious circle. When the only or main response to aggression is repression, it may increase physical/psychic threats and stress perceived by the aggressor, and in turn aggressive behaviors. If we respond to the “act of aggression” with only an “act of repression”, we enter into an escalation of events, which, sooner or later (after the effect of the medication has worn off or the jail term has been served), will lead to a reinforcement of aggressive behaviors.

The existence of several levels of understanding gives access to several levels of intervention. Interventions can occur at the level of stressful situations (level 1), at the level of the aggressor (perception of the stressful situation: level 2a, and response: level 2b), and at the level of the aggressed individual (perception of the aggression: level 3a, and response: level 3b).

4.1. Level of the stressful situation

We can try to alleviate the stressful situation (see figure level 1) or in extreme cases avoid it. It is important to intervene early if a growing child’s environment is hostile. This hostile environment exposes the child to a chronic stress situation leading to aggressive behaviors for surviving and for asserting his/her existence. The child’s experience of familial or social violence, with emotional deficit, influences his or her perception of aggressiveness (both his or her own aggressiveness and that of others) and leads to maladaptive aggressive behaviors. The very notion of “adaptive” or “maladaptive” aggression in fact depends on the threshold of familial or social tolerance for aggressiveness. Patterson [79] suggests that aggressiveness in children is related to a parental style of high tolerance of aggressive behavior. A violent familial or social environment can lead a child to repeated aggressive behaviors. This behavior repeats itself because, for the child, aggression is the only known response to a stressful situation. Also, repetition has the value of pain reactivation, of confirming the familial or social belonging, of reaffirming the individual’s origins.

In addition, there is evidence that parental neglect and physical, sexual or emotional abuse can lead children to poor peer relations

and aggression [80–82]. Findings from a cohort study conducted on 908 individuals who experienced child abuse and neglect and on 667 controls, provide strong support for the cycle of violence hypothesis: being abused or neglected as a child increases risk for violent behavior, delinquency, and adult criminal behavior [83].

We cannot totally “control” a hostile environment. Nor can we eliminate all stressful situations. It is not even clear that this would be desirable. It is easier to intervene at the level of the “aggressor” and the “aggressed” individual in their perception of situations (stressful situations, situations of aggression) and their responses.

4.2. Level of the aggressor’s perception of the stressful situation

We could try to intervene in the aggressor’s perception and interpretation of stressful situations by social learning (level 2a). This social learning can include children at risk and be preventive (by using education, media, etc.) or it can involve strategies to help the aggressor to perceive and interpret the information differently. A situation seen previously as threatening might be then interpreted as non-dangerous and non-stressful. Some studies [84] suggest that one of the most robust indicators of aggressive behavior in juvenile delinquents is a tendency toward paranoia; these children have a tendency to attribute erroneously hostile intent to others. Aggressive behaviors could result from maladaptive responses to a social situation due to a learning deficit or a deficiency in the information encoding process [85,86]. According to Vitiello and Stoff [1], aggressive children of the impulsive-reactive-hostile-affective type can be considered a priori to be less capable of self-control and to be driven to aggression by poor frustration tolerance, negative affects such as anger and fear, and a cognitive distortion of the environmental circumstances. Therefore, interventions aimed at preventing, decreasing, or counteracting a high level of arousal, on the one hand, and at improving cognitive functioning, on the other, can have a therapeutic role.

4.3. Level of the aggressor’s response to the stressful situation

We can modify the response of the aggressor to the stressful situation (level 2b) by helping him/her to cope with stress differently, to control better impulsivity, and to find responses other than aggressive behavior resulting from a conflictual situation. Karli [87] insists on the main role of emotions and affective experience in aggression; according to this author, emotion would provide the primary energy for aggression. Therefore, we can focus on the development of strategies based on body work through body-mind therapy, such as hypnosis, for releasing stress-induced tension with

other ways than aggression and holding better stress and negative emotions like fear or anger without explosive crises.

4.4. Level of the aggressed individual's perception of the aggressive behavior

The aggressed individual could also perceive aggressive behaviors differently (level 3a). We observe a greater tolerance of aggression when there is a better understanding of aggressive behavior. This is what was described in the example of the aggression curve, which enabled a clinical team to understand and manage better the aggression in children with autism. To improve understanding of aggressive behaviors through their evaluation and create a space for thinking and keeping at distance negative emotions (such as fear), the Other-Injurious Behavior Scale (OIBS) is very useful [60,61]. It provides a specific validated tool to assess quantitatively and qualitatively aggressive behavior and concurrent associated behaviors in the aggressor (see Section 2.3.2), but also the reaction and responses to aggression in the aggressed individual. The OIBS situates the aggressive behavior to be evaluated within a dynamic of interaction/communication between an individual who expresses him/herself through OIB and an aggressed individual who reacts to it.

Lack of understanding is a source of fear — fear of the unknown, of the absence of limits, of that which seems uncontrollable. This fear explains the necessity of “controlling” the aggressor by means of prolonged chemical, physical or legal repression. But the aggressed individual could also interpret the aggression differently and relativize the notion of danger (when, of course, this is realistic). That which is perceived as a dangerous aggressive behavior within one social group can be perceived differently in another. Also, the very perception of danger can decrease if the aggressed individual learns to protect himself/herself (e.g. by means of self-defense). Knowing how to deal better with aggression, he/she feels less threatened, is less fearful, and thus can look for alternative solutions other than repression such as decreasing the stressful situation experienced by the aggressor.

4.5. Level of the aggressed individual's response to the aggressive behavior

In the end, it is possible to respond to aggressive behaviors using means other than repression (level 3b). If we want to escape the vicious circle described in the figure, it is essential to stop responding to the “aggressive act” by the “repressive act”. The aggressed individual could learn to verbalize his or her fear and to communicate with the aggressor, while (obviously) still protecting himself/herself. It is not a matter of communicating with the aggressor during the fit. At the very moment of aggression, the aggressor is acting rather than listening. It is therefore a matter of trying to understand and communicate in a calm spell and at distance from the aggression.

We may ask ourselves why we do not tackle stress or aggression head-on. It appears indeed necessary not to simply suppress stress or aggressive behavior. The aggressive individual perceives or even defines himself or herself as being stressed and aggressive. He/she expresses him/herself and exists through his/her stress and through his or her identity as an aggressor. If we administer consistently medications, which prevent the aggressor individual from externalizing his/her stress or aggressiveness, we prevent also him/her from expressing himself/herself. In making the stress or aggression disappear, we make also the aggressor disappearing. This is equivalent to negating the person, to negating what he or she is. We are attempting to annihilate the aggressive person through the very things, which define this person. This can only

represent a psychic threat generating stress, to which the aggressor will respond (when possible) with more aggressive behaviors.

In the event of violent behavior, prescribing temporary isolation and sedatives can be effective both in soothing the patient and in helping him/her to stop the acting out. But once the “storm” has passed, it is important to initiate a dialogue, to introduce the spoken words, all the while recognizing the individual in his/her stress and aggressiveness. Communication can thus be established and work can begin. As long as we look to eliminate aggressive behavior (by using chronic medication) or even the aggressive individual from society (with prison or long-term isolation in a psychiatric hospital or special educative center), the protection of others will be assured only momentarily. In reality, and in the long run, nothing will be solved. In reacting this way, we are putting ourselves in a purely negative approach: we are “rubbing out” a behavior or a person. Furthermore, studies in animal models show clearly that social isolation stress induces aggressive behavior [36–39]. Interestingly, some recent studies in humans have reported strong associations between perceived loneliness and aggression [88,89]. The Ayesha et al. study [88] conducted on 120 boys and 120 girls living in institutionalized care showed that the adolescents perceiving a higher level of loneliness expressed more physical aggression. Similarly, the Yavuzer et al. study [89] conducted on a non-clinical population of 904 students showed that perceived loneliness predicted significantly aggression. Finally, the exclusion of violent adolescents from society by regrouping them in jail or special educative centers can also lead to a mutual reinforcement of aggression (by group effect and they learn from more violent and experienced peers), and allow them to find their place and identity within a new group, the one of violent adolescents. The dissuasive role of a transitory exclusion is not anymore operating. In this context, exclusion does not represent anymore a possible mobilizing experience. Exclusion is then only reduced to a role of group protection and repression.

Trying to understand the act of aggression is in itself proof that we no longer attempt to erase it. It is a matter of recognizing the individual who expresses himself/herself through this behavior. Understanding and developing adapted responses to this understanding, for the individual and the society, is a far more positive and constructive enterprise.

5. Conclusion

No model can possibly account for all the factors and mechanisms involved in aggression. An integrated psychological, sociological, and environmental approach would allow us to understand better aggressive behaviors. A physiological approach studying biological factors playing a role in aggression, notably biological stress responses, was not discussed here, but is equally important.

If there was one main idea to be retained from this article, one word, it would be the word *understanding*. We must seek to understand the different links that form the “chain of aggressive behavior” (particularly, the impact of the response of the aggressed individual on the aggressor with a risk for reinforcement of aggression creating a vicious circle). It is important to understand the complexity of aggressive behavior, not to reduce this behavior to a simple symptom whose solution is either medication or repression. There exist several levels of intervention, levels which are not limited, as it is most often the case, to the acted out response of the aggressed individual or society. Also, we have to recognize the difficulty of a direct, short-term intervention for getting out of this vicious circle, especially because the various factors involved in aggression interact among themselves. The more clinicians are aware of the complexity of aggressive behavior in a child or adoles-

cent, the more possibilities become available to help this individual and his/her family.

Finally, if the model proposed here allows us a better understanding, even small, of aggressive behavior, by thinking about aggressive behavior in a more global and interactive approach, this model will be already useful.

Disclosure of interest

The author declares that she has no competing interest.

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